

HDOC Day-33

Associative Property of Matrix Addition and Multiplication in C

Problem Statement

Create a menu-driven C program using separate functions to verify $(A+B)+C = A+(B+C)$ and $(A \times B) \times C = A \times (B \times C)$. Use variable-length 2D integer arrays and display the matrices obtained on both sides.

Algorithm

1. Display menu repeatedly until Exit is selected.
2. Choice 1: read positive m and n; read A, B and C of size m x n; calculate $(A+B)+C$ and $A+(B+C)$; display and compare both matrices.
3. Choice 2: read positive m, n, p and q; read A(m x n), B(n x p), C(p x q); calculate AB, BC, $(AB)C$ and $A(BC)$; display and compare both matrices.
4. Choice 3: exit. For another choice, print an error message.

Test Case Table

TC	Choice	Input	Expected Result
1	1	m=2,n=2; A=1 2/3 4; B=5 6/7 8; C=1 1/1 1	Both = 7 9/11 13; addition verified
2	1	m=1,n=3; A=-1 0 2; B=3 -2 1; C=4 5 -3	Both = 6 3 0; addition verified
3	2	m=n=p=q=2; A=1 2/3 4; B=5 6/7 8; C=identity	Both = 19 22/43 50; multiplication verified
4	2	m=1,n=2,p=2,q=1; A=1 2; B=3 4/5 6; C=2/1	Both = 40; multiplication verified
5	1	Enter m=0, then m=2 and valid remaining input	Rejects 0 and asks again
6	2	Enter q=-1, then q=2 and valid matrices	Rejects -1 and asks again
7	9	Invalid menu choice	Invalid choice. Please enter 1 to 3.
8	3	Exit	Exiting program.

C Program

```
#include <stdio.h>

void associativeAddition(void);
void associativeMultiplication(void);

int main(void) {
    int choice;
    do {
        printf("\n--- MATRIX ASSOCIATIVE PROPERTY MENU ---\n");
        printf("1. Verify associative property of matrix addition\n");
        printf("2. Verify associative property of matrix multiplication\n");
        printf("3. Exit\nEnter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1: associativeAddition(); break;
            case 2: associativeMultiplication(); break;
```

```

        case 3: printf("Exiting program.\n"); break;
        default: printf("Invalid choice. Please enter 1 to 3.\n");
    }
} while (choice != 3);
return 0;
}

void associativeAddition(void) {
    int m,n,i,j,verified=1;
    do { printf("Enter number of rows m: "); scanf("%d",&m); if(m<=0) printf("m must be positive.\n"); }
while(m<=0);
    do { printf("Enter number of columns n: "); scanf("%d",&n); if(n<=0) printf("n must be positive.\n"); }
while(n<=0);
    int A[m][n],B[m][n],C[m][n],AB[m][n],BC[m][n],left[m][n],right[m][n];
    printf("Enter elements of matrix A:\n"); for(i=0;i<m;i++) for(j=0;j<n;j++) scanf("%d",&A[i][j]);
    printf("Enter elements of matrix B:\n"); for(i=0;i<m;i++) for(j=0;j<n;j++) scanf("%d",&B[i][j]);
    printf("Enter elements of matrix C:\n"); for(i=0;i<m;i++) for(j=0;j<n;j++) scanf("%d",&C[i][j]);
    for(i=0;i<m;i++) for(j=0;j<n;j++){ AB[i][j]=A[i][j]+B[i][j]; BC[i][j]=B[i][j]+C[i][j];
left[i][j]=AB[i][j]+C[i][j]; right[i][j]=A[i][j]+BC[i][j]; }
    printf("\n(A+B)+C:\n"); for(i=0;i<m;i++){ for(j=0;j<n;j++) printf("%d ",left[i][j]); printf("\n"); }
    printf("\nA+(B+C):\n"); for(i=0;i<m;i++){ for(j=0;j<n;j++){ printf("%d ",right[i][j]);
if(left[i][j]!=right[i][j]) verified=0; } printf("\n"); }
    printf(verified ? "Associative property of matrix addition is verified.\n" : "Associative property of
matrix addition is not verified.\n");
}

void associativeMultiplication(void) {
    int m,n,p,q,i,j,k,verified=1;
    do { printf("Enter m: "); scanf("%d",&m); if(m<=0) printf("m must be positive.\n"); } while(m<=0);
    do { printf("Enter n: "); scanf("%d",&n); if(n<=0) printf("n must be positive.\n"); } while(n<=0);
    do { printf("Enter p: "); scanf("%d",&p); if(p<=0) printf("p must be positive.\n"); } while(p<=0);
    do { printf("Enter q: "); scanf("%d",&q); if(q<=0) printf("q must be positive.\n"); } while(q<=0);
    int A[m][n],B[n][p],C[p][q],AB[m][p],BC[n][q],left[m][q],right[m][q];
    printf("Enter elements of matrix A (%d x %d):\n",m,n); for(i=0;i<m;i++) for(j=0;j<n;j++)
scanf("%d",&A[i][j]);
    printf("Enter elements of matrix B (%d x %d):\n",n,p); for(i=0;i<n;i++) for(j=0;j<p;j++)
scanf("%d",&B[i][j]);
    printf("Enter elements of matrix C (%d x %d):\n",p,q); for(i=0;i<p;i++) for(j=0;j<q;j++)
scanf("%d",&C[i][j]);
    for(i=0;i<m;i++) for(j=0;j<p;j++){ AB[i][j]=0; for(k=0;k<n;k++) AB[i][j]+=A[i][k]*B[k][j]; }
    for(i=0;i<n;i++) for(j=0;j<q;j++){ BC[i][j]=0; for(k=0;k<p;k++) BC[i][j]+=B[i][k]*C[k][j]; }
    for(i=0;i<m;i++) for(j=0;j<q;j++){ left[i][j]=0; for(k=0;k<p;k++) left[i][j]+=AB[i][k]*C[k][j]; }
    for(i=0;i<m;i++) for(j=0;j<q;j++){ right[i][j]=0; for(k=0;k<n;k++) right[i][j]+=A[i][k]*BC[k][j]; }
    printf("\n(A x B) x C:\n"); for(i=0;i<m;i++){ for(j=0;j<q;j++) printf("%d ",left[i][j]); printf("\n");
}
    printf("\nA x (B x C):\n"); for(i=0;i<m;i++){ for(j=0;j<q;j++){ printf("%d ",right[i][j]);
if(left[i][j]!=right[i][j]) verified=0; } printf("\n"); }
    printf(verified ? "Associative property of matrix multiplication is verified.\n" : "Associative
property of matrix multiplication is not verified.\n");
}

```

Compilation

```
gcc -std=c99 hdoc33.c -o hdoc33
./hdoc33
```

Screenshot Spaces

Screenshot 1 - successful compilation

```

(kali㉿kali)-[~/Desktop]
$ vi hdoc33.c

(kali㉿kali)-[~/Desktop]
$ gcc hdoc33.c -o hdoc33 -lm

(kali㉿kali)-[~/Desktop]
$ ./hdoc33

```

Screenshot 2 - Choice 1: Associative addition

```

— MATRIX ASSOCIATIVE PROPERTY MENU —
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 1
Enter number of rows m: 2
Enter number of columns n: 2
Enter elements of matrix A:
1 2
3 4
Enter elements of matrix B:
5 6
7 8
Enter elements of matrix C:
1 1
1 1
(A+B)+C:
7 9
11 13
A+(B+C):
7 9
11 13
Associative property of matrix addition is verified.

```

```

— MATRIX ASSOCIATIVE PROPERTY MENU —
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 1
Enter number of rows m: 1
Enter number of columns n: 3
Enter elements of matrix A:
-1 0 2
Enter elements of matrix B:
3 -2 1
Enter elements of matrix C:
4 5 -3
(A+B)+C:
6 3 0
A+(B+C):
6 3 0
Associative property of matrix addition is verified.

```

```

— MATRIX ASSOCIATIVE PROPERTY MENU —
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 1
Enter number of rows m: 0
m must be positive.
Enter number of rows m: 2
Enter number of columns n: 0
n must be positive.
Enter number of columns n: 2
Enter elements of matrix A:
1 2
3 4
Enter elements of matrix B:
5 6
7 8
Enter elements of matrix C:
9 10
11 12

(A+B)+C:
15 18
21 24

A+(B+C):
15 18
21 24
Associative property of matrix addition is verified.

```

Screenshot 3 - Choice 2: Associative multiplication

```

— MATRIX ASSOCIATIVE PROPERTY MENU —
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 2
Enter m: 2
Enter n: 2
Enter p: 2
Enter q: 2
Enter elements of matrix A (2 x 2):
1 2 3 4
Enter elements of matrix B (2 x 2):
4 6 7 8
Enter elements of matrix C (2 x 2):
1 0
0 1

(A x B) x C:
18 22
40 50

A x (B x C):
18 22
40 50
Associative property of matrix multiplication is verified.

```

```

— MATRIX ASSOCIATIVE PROPERTY MENU —
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 2
Enter m: 1
Enter n: 2
Enter p: 2
Enter q: 1
Enter elements of matrix A (1 x 2):
1 2
Enter elements of matrix B (2 x 2):
3 4
5 6
Enter elements of matrix C (2 x 1):
2
1
(A x B) x C:
42
A x (B x C):
42
Associative property of matrix multiplication is verified.

```

```

— MATRIX ASSOCIATIVE PROPERTY MENU —
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 2
Enter m: 1
Enter n: 1
Enter p: 1
Enter q: -1
q must be positive.
Enter q: 2
Enter elements of matrix A (1 x 1):
2
Enter elements of matrix B (1 x 1):
3
Enter elements of matrix C (1 x 2):
4 5
(A x B) x C:
24 30
A x (B x C):
24 30
Associative property of matrix multiplication is verified.

```

Screenshot 4 - Validation / Exit

```

— MATRIX ASSOCIATIVE PROPERTY MENU —
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 9
Invalid choice. Please enter 1 to 3.

```

```

— MATRIX ASSOCIATIVE PROPERTY MENU —
1. Verify associative property of matrix addition
2. Verify associative property of matrix multiplication
3. Exit
Enter your choice: 3
Exiting program.

```